

RAM and Storage Assignment

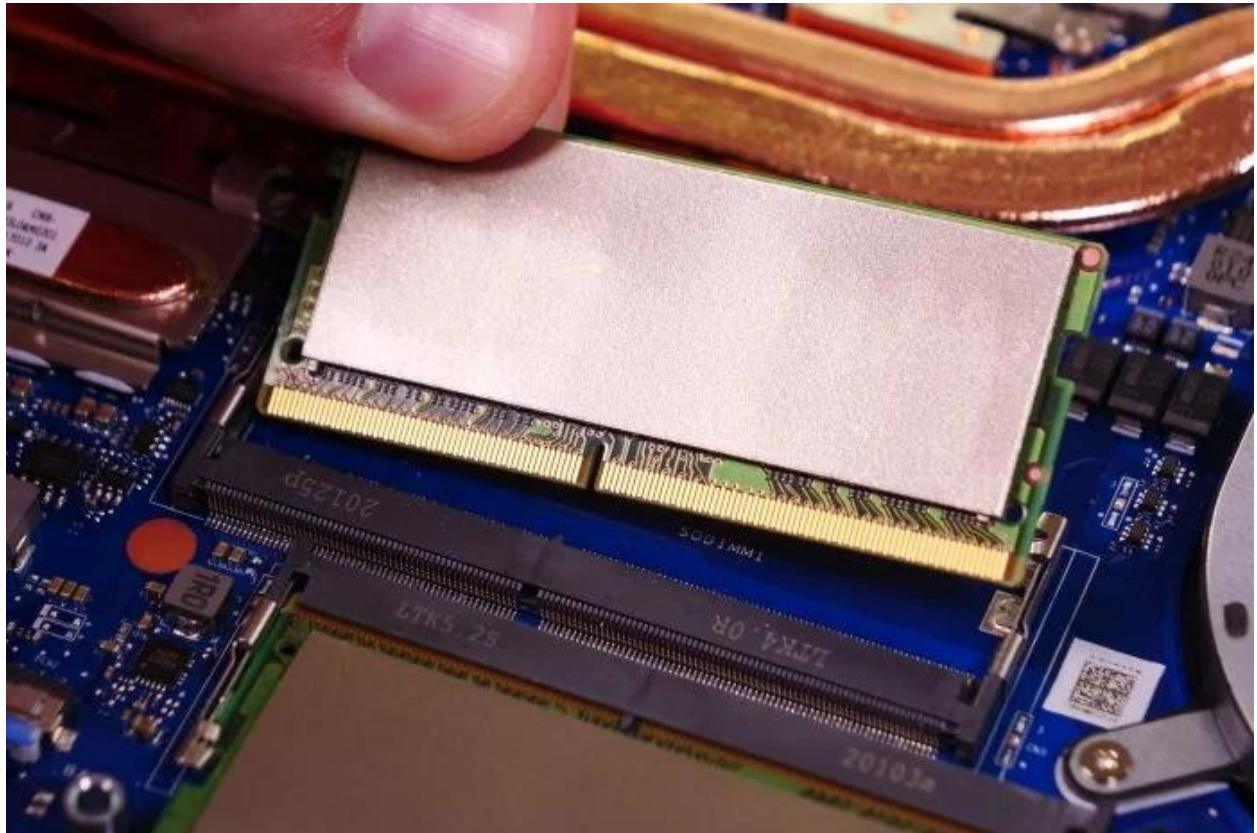
1. DDR3 vs DDR4 vs DDR5 RAM Comparison

Feature	DDR3	DDR4	DDR5
Release Year	2007	2014	2020
Typical Speed	800–2133 MT/s	2133–3200 MT/s	4800–8400+ MT/s
Voltage	1.5V	1.2V	1.1V
Bandwidth	Lower	Higher than DDR3	Highest
Power Efficiency	Lowest	Better	Best
Capacity per Module	Up to 16 GB	Up to 64 GB	128 GB+ possible
Typical Use Cases	Older PCs, office computers	Modern desktops, gaming PCs	High-end gaming, AI, content creation

Summary

- **DDR3** is older and slower.
- **DDR4** is the most common in many current PCs.
- **DDR5** provides the highest performance and efficiency for modern systems.

2. RAM Removal and Reinstallation (Practical)



3. HDD vs SSD vs NVMe Comparison

Feature	HDD	SATA SSD	NVMe SSD
Technology	Spinning disks	Flash memory	Flash memory via PCIe
Typical Speed	80–160 MB/s	500–550 MB/s	2000–7000+ MB/s
Noise	Audible	Silent	Silent
Durability	Moving parts	No moving parts	No moving parts
Cost	Cheapest	Moderate	Most expensive
Best For	Large storage	Everyday computing	Gaming and professional workloads

Example Use Cases

Storage Type	Example App/Use Case
HDD	Movies, photos, document archives
SSD	Microsoft Office, web browsing, Zoom
NVMe SSD	FIFA, PUBG, Adobe Premiere Pro, Photoshop

Three Key Differences

1. **Speed:** NVMe > SSD > HDD.
2. **Durability:** SSD and NVMe have no moving parts, making them more resistant to shock.
3. **Connection Method:** HDD/SSD commonly use SATA, while NVMe uses PCIe through an M.2 slot.

4. Storage Drive Removal and Reinstallation

https://youtube.com/shorts/pB_qW0YLiTE?si=1UZkmFxmEmjAr3zG